

Jacobson Rings and Zariski's Lemma

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Introduction

This note is a synthesis of my favorite Nullstellensatz proofs that I have read. It proves the Nullstellensatz as a corollary of deep results on Jacobson rings. In particular, I will show that the Jacobson property is in a surprisingly strict sense equivalent to the conditions needed for the Nullstellensatz (minus the algebraically closed property). This causes all the Nullstellensätze (weak, strong, etc) to drop out almost trivially once we understand the Jacobson condition. While reading the note, when I write "Jacobson ring" you should think of $k[x_1, \dots, x_n]/I$. Upon reading each theorem think about how it corresponds to a property of this ring, as well as how each result is related to the Nullstellensatz and algebraic geometry as you understand it (the naming should help).

The background required is a solid knowledge of integral ring extensions. For instance by reading the fifth chapter of Atiyah Macdonald.

The Proof

Definition 2.1 (Jacobson Ring). A ring is Jacobson if it satisfies one of the following equivalent conditions.

- (a) Every prime ideal is the intersection of the maximal ideals containing it.
- (b) Every radical ideal is the intersection of the maximal ideals containing it.
- (c) In every quotient ring the nilradical is equal to the Jacobson radical.

Proof. Take (a) as the definition. The Jacobson condition essentially says that any intersection of prime ideals can be written as an intersection of maximal ideals:

$$\bigcap_i \mathfrak{p}_i = \bigcap_i \bigcap_{\mathfrak{p}_i \subseteq \mathfrak{m}} \mathfrak{m}$$

In particular the intersection of the prime ideals over an ideal I is equal to the intersection of the maximal ideals over I . With this in mind we prove that (b) and (c) are equivalent to (a):

- (b) A radical ideal is characterized as an intersection of prime ideals, therefore apply (a). Furthermore (a) is a special case since prime ideals are radical.
- (c) By the lattice theorem the nilradical (resp Jacobson radical) of R/I corresponds to the intersection of the prime ideals (resp maximal ideals) containing I , therefore apply (a). Furthermore (a) is a special case when we quotient by a prime ideal.

□

Lemma 2.2 (Rabinowitsch Criterion). R is a Jacobson ring if and only if for every $s \in R$ the natural map

$$\text{Spec}(R[s^{-1}]) \rightarrow \text{Spec}(R)$$

sends maximal ideals to maximal ideals.

Proof. Recall that

$$\text{Spec}(R[s^{-1}]) \cong \{\mathfrak{p} \in \text{Spec}(R) \mid s \notin \mathfrak{p}\}$$

which implies that the maximal ideals of $\text{Spec}(R[s^{-1}])$ correspond to the ideals which are maximal among those which do not contain s .

$\Rightarrow$ Suppose that $\mathfrak{p} \in \text{Spec}(R)$ is maximal in this sense. Then either $\mathfrak{p}$ is maximal or every maximal ideal containing it contains s . The second case implies that $\mathfrak{p}$ contains s by the Jacobson property, a contradiction. Therefore all such ideals are maximal in R .

$\Leftarrow$ Suppose that $\mathfrak{p}$ is a prime ideal of R and $s \notin \mathfrak{p}$. Then localize at s and note that $\mathfrak{p}R[s^{-1}]$ is contained in some maximal ideal $\mathfrak{n} \in \text{Spm}(R[s^{-1}])$. Then $\mathfrak{n}$ corresponds to a maximal $\mathfrak{n}' \in \text{Spm}(R)$ containing $\mathfrak{p}$ but not s . Therefore $s \notin \bigcap_{\mathfrak{p} \subseteq \mathfrak{m}} \mathfrak{m}$ and $\mathfrak{p} = \bigcap_{\mathfrak{p} \subseteq \mathfrak{m}} \mathfrak{m}$.

□

Another interpretation of the Rabinowitsch criterion is that R is Jacobson if and only if $(R/\mathfrak{p})[s^{-1}]$ being a field implies that $R/\mathfrak{p}$ is a field. This is because the natural map will send $\mathfrak{p}' \in \text{Spm}(R[s^{-1}])$ to a maximal ideal in R . It is worth noting that the Rabinowitsch criterion is some kind of abstract reformulation of its namesake "trick". For a proof relying on that trick check out [4].

Lemma 2.3. Quotients and single element localizations of Jacobson rings are Jacobson.

Proof. For quotients, apply characterization (c) in Definition 2.1, noting that $(R/I)/J \cong R/\pi^{-1}(J)$ where $\pi^{-1}(J)$ is the lift of J along the quotient map.

For localizations we will show that the Rabinowitsch criterion is satisfied on $R[s^{-1}]$. Suppose that $b = \frac{a}{s^k} \in R[s^{-1}]$ and note that $R[s^{-1}][b^{-1}] \cong R[(as)^{-1}]$. Therefore consider the natural maps

$$\begin{array}{ccccc} \text{Spec}(R[s^{-1}][b^{-1}]) & \longrightarrow & \text{Spec}(R[s^{-1}]) & \longrightarrow & \text{Spec}(R) \\ & & \parallel & \nearrow & \\ & & \text{Spec}(R[(as)^{-1}]) & & \end{array}$$

and note that the diagonal arrow sends maximal ideal to maximal ideals since R is Jacobson. Therefore both the horizontal arrows send maximal ideals to maximal ideals and the Rabinowitsch criterion is satisfied on $R[s^{-1}]$. □

Lemma 2.4. Let $A \subset B$ be integral domains, B finite-type over A . Then for any nonzero $v \in B$ there exists a nonzero $u \in A$ so that every ring homomorphism $\phi : A \rightarrow \Omega$ to an algebraically closed field with $\phi(u) \neq 0$ can be extended to $\tilde{\phi} : B \rightarrow \Omega$ with $\tilde{\phi}(v) \neq 0$.

Proof. By induction we reduce to the case where B is generated by 1 element over A , call this element x . There are two cases

Case 1: x is transcendental over A . Let $v = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ and take $u = a_n$. We choose some $\tilde{\phi}(x) = \zeta$ which does not satisfy the image of this polynomial in Ω .

Case 2: x is algebraic over A . Write

$$a_m x^m + a_{m-1} x^{m-1} + \cdots + a_0 = 0 \quad b_n v^n + b_{n-1} v^{n-1} + \cdots + b_0 = 0$$

and let $u = a_m b_0$. Then in the ring $B[u^{-1}]$ we can write

$$x^m + \frac{a_{m-1}}{a_m} x^{m-1} + \cdots + \frac{a_0}{a_m} = 0 \quad v^{-1} = -\frac{1}{b_0} (b_n v^{n-1} + b_{n-1} v^{n-2} + \cdots + b_1)$$

so that $B[u^{-1}]$ is finite over $A[u^{-1}]$ (since there is a monic polynomial for x) and v becomes a unit. Extend ϕ to a homomorphism $\phi' : A[u^{-1}] \rightarrow \Omega$, let $\mathfrak{p} = \ker \phi'$, and apply lying over to find some $\mathfrak{q} \in \text{Spec}(B[u^{-1}])$ lying over $\mathfrak{p}$. Then use the property of algebraically closed fields to complete the diagram of maps:

$$\begin{array}{ccccccc} A & \hookrightarrow & A[u^{-1}] & \longrightarrow & A[u^{-1}]/\mathfrak{p} & \hookrightarrow & \Omega \\ \downarrow & & \downarrow & & \downarrow & \nearrow & \\ B & \hookrightarrow & B[u^{-1}] & \longrightarrow & B[u^{-1}]/\mathfrak{q} & & \end{array}$$

Since v is a unit in $B[u^{-1}]$ it is not contained in $\mathfrak{q}$ and not killed by the map $B \rightarrow \Omega$ as desired. $\square$

Theorem 2.5 (Generalized Zariski Lemma). R is a Jacobson ring if and only if every finite-type R -algebra that is a field is finite over R .

Proof.

$\Rightarrow$ Since quotients are Jacobson we may assume S contains R as a subring, therefore R is an integral domain. We apply Lemma 2.4 on $R \subset S$ and $v = 1$ to get u with the stated property. Since (0) is prime in R take $\mathfrak{m}$ to be a maximal ideal not containing u and Ω to be the algebraic closure of $R/\mathfrak{m}$. The map $R \rightarrow R/\mathfrak{m} \rightarrow \Omega$ does not kill u and therefore extends to an embedding $\tilde{\phi} : S \rightarrow \Omega$. Therefore S is algebraic over $R/\mathfrak{m}$, hence integral because $R/\mathfrak{m}$ is a field. Since integral finite-type algebras are finite, S is finite over $R/\mathfrak{m}$ and by extension R .

$\Leftarrow$ Suppose that $\mathfrak{p} \in \text{Spec}(R)$ and $f \notin \mathfrak{p}$. We need to find a maximal ideal containing $\mathfrak{p}$ but not f . Let $S = R/\mathfrak{p}$ and let $\mathfrak{m}$ be a maximal ideal of $S[f^{-1}]$. Then $S[f^{-1}]/\mathfrak{m}$ is finite over R (and therefore $S/(\mathfrak{m} \cap S)$) by assumption. By integrality $\mathfrak{m} \cap S$ is maximal in S and by construction corresponds to an ideal containing $\mathfrak{p}$ but not containing f . $\square$

As a fun remark, we can collect all these results into a very satisfying collection of equivalent conditions for a ring to be Jacobson:

- (a) Every prime ideal is the intersection of the maximal ideals containing it.
- (b) Every radical ideal is the intersection of the maximal ideals containing it.
- (c) In every quotient ring the nilradical is equal to the Jacobson radical.
- (d) For every $s \in R$ the natural map $\text{Spec}(R[s^{-1}]) \rightarrow \text{Spec}(R)$ sends maximal ideals to maximal ideals.
- (e) If $\mathfrak{p} \in \text{Spec}(R)$ and $s \in R$ then $(R/\mathfrak{p})[s^{-1}]$ being a field implies that $R/\mathfrak{p}$ is a field.
- (f) Every finite-type R -algebra that is a field is finite over R .

Corollary 2.6 (Generalized Nullstellensatz). If R is Jacobson and S is a finite-type R -algebra, then S is Jacobson.

Proof. We will use the characterization of Jacobson rings given in Theorem 2.5. Suppose K is a field which is finite-type over S . Then by transitivity it is finite-type over R and therefore finite over R . Therefore it is finite over S . $\square$

Theorem 2.7 (Weak Nullstellensatz). Maximal ideals of $k[x_1, \dots, x_n]$ correspond to finite extensions of k . Moreover, if k is algebraically closed then the points of $\mathbb{A}^n$ are in bijective correspondence with the maximal ideals of $k[x_1, \dots, x_n]$.

Proof. Since k is Jacobson $k[x_1, \dots, x_n]/\mathfrak{m}$ is finite over k by Theorem 2.5, hence a finite extension of k . Now suppose that k is algebraically closed. Then there are no nontrivial finite extensions of k and $k[x_1, \dots, x_n]/\mathfrak{m} \cong k$. In particular the quotient map corresponds to evaluation at the point $(\text{im}(x_1), \dots, \text{im}(x_n))$ and $\mathfrak{m}$ is uniquely characterized as the kernel of this evaluation. $\square$

Theorem 2.8 (Strong Nullstellensatz). If k is algebraically closed and I is an ideal of $k[x_1, \dots, x_n]$ then:

$$\mathcal{I}(\mathcal{V}(I)) = \sqrt{I}$$

In particular, the affine algebraic sets (resp affine varieties) are in bijective correspondence with the radical ideals (resp prime ideals) of $k[x_1, \dots, x_n]$.

Proof. Note that

$$\sqrt{I} = \bigcap_{I \subseteq \mathfrak{p}} \mathfrak{p} \quad \underset{\text{(Jacobson Property)}}{=} \quad \bigcap_{I \subseteq \mathfrak{m}} \mathfrak{m} \quad \underset{\text{(Weak N-satz)}}{=} \quad \bigcap_{P \in \mathcal{V}(I)} \mathfrak{m}_P \quad \underset{\text{(Def)}}{=} \quad \mathcal{I}(\mathcal{V}(I))$$

where we have used $\mathfrak{m}_P = \{f | f(P) = 0\}$ to denote the maximal ideal defined by evaluation at P . $\square$

References

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